Quantum Interference Transformer (QIT): Emergent Sequence Intelligence from Amplitude Dynamics

Mihail Stancescu
Independent Research
madmishu007@gmail.com

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Abstract

We introduce the **Quantum Interference Transformer** (QIT), a sequence model in which token relationships are resolved not by dot-product attention scores, but by constructive and destructive interference within a parameterised quantum circuit. The central claim is modest but concrete: *sequence intelligence can emerge from interference dynamics*, and an interference-based attention mechanism is measurably more sample-efficient than classical attention on structured tasks that have a natural phase-kickback representation.

We present **QIT-0**, a minimal prototype built on PennyLane [3] and PyTorch. The architecture encodes a token sequence into qubit rotation angles, seeds cross-token correlations via a ring CNOT ladder, applies a learned strongly-entangling unitary as the attention operator, and reads out Pauli-Z expectation values as the hidden state. A single linear layer decodes these to class logits.

On the 4-bit binary parity task — a memorisation benchmark over all 16 inputs — QIT-0 (78 parameters) reaches 100% accuracy in $\mathbf{6.2 \pm 1.7}$ **epochs** (mean ± std over 5 random seeds, all 5 converging). An MLP (94 parameters) requires 31.7 ± 4.1 epochs (3 / 5 seeds converge in 200 epochs), a GRU (110 parameters) requires 34.5 ± 5.5 epochs (2 / 5 converge), and a classical Transformer [18] (206 parameters) fails to converge in 200 epochs under any of the 5 seeds tested. A logistic regression baseline on the sum-feature representation $f(\mathbf{x}) = \sum_i x_i$ also fails to converge — parity is not linearly separable even with this structural feature, as the decision boundary is non-monotone in the sum. Wall-clock time per epoch is ≈195 ms due to quantum simulation overhead, compared to sub-2 ms for classical models; this cost is intrinsic to software simulation and is not representative of future hardware execution.

We do not claim that quantum computation is generally superior to classical computation for machine learning. We claim that for tasks whose structure maps naturally to quantum phase dynamics, interference-based attention exhibits strong inductive bias and superior sample efficiency.

Contents

1 Introduction

The dominant paradigm for sequence modelling is softmax attention [18], in which relevance between token pairs is computed as an explicit dot-product similarity followed by a normalised weighted sum of values. This mechanism is expressive and well-understood, but it imposes a particular inductive bias: relevance is a scalar pairwise comparison, and aggregation is a convex combination. Whether this is the *only* useful inductive bias for sequence tasks is an open question.

Quantum mechanics offers an alternative picture of how information from multiple sources can be combined. In quantum circuits, information propagates as complex amplitudes, and multiple signal paths interfere. Paths with aligned phases amplify each other (*constructive* interference); paths with opposing phases cancel (*destructive* interference). The result of a computation is determined not by explicit similarity scoring but by the global amplitude pattern that emerges from the circuit geometry and the learned unitary parameters. Critically, this mechanism is *non-local by construction*: a single layer of entangling gates connects every qubit to every other, giving the circuit access to all token positions simultaneously.

This paper asks a simple question: *can this interference dynamic serve as a drop-in replacement for dot-product attention?*

To answer it, we build QIT-0, the first prototype of the Quantum Interference Transformer. QIT-0 is explicitly a proof-of-concept, not a production system. The circuit runs on a classical quantum simulator (PennyLane's `default.qubit`), which makes wall-clock time uncompetitive with classical models. The vocabulary is binary (2 tokens), the sequence length is 4, and the only task is parity. But the results are striking: the interference mechanism finds the parity function in 12.4 gradient steps (mean, 5 seeds), while the classical Transformer with 2.6× more parameters fails to converge in 400 gradient steps (200 epochs) across all seeds tested.

The paper is organised as follows. Section 2 reviews the relevant background. Section 3 describes the QIT architecture in full. Section 4 describes the experimental setup. Section 5 presents results. Section 6 discusses what the results mean and what they do not mean. Section 7 describes planned future work. Section 8 concludes.

2 Background

2.1 Classical Attention

The scaled dot-product attention mechanism introduced by Vaswani et al. [18] computes, for a sequence of token representations $X \in \mathbb{R}^{n \times d}$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{1}$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$ are learned linear projections. The softmax produces a probability distribution over positions, used to form a weighted sum of value vectors. Relevance between tokens is an explicit bilinear function of their representations.

This mechanism is powerful but has several well-known properties. The quadratic cost in sequence length $O(n^2 d)$ limits scalability. The inductive bias — similarity as a dot product — is appropriate for many tasks but is not a universal prior. Even a minimal single-head Transformer for 4-token binary sequences requires 206 parameters in our experiments.

2.2 Quantum Computing Fundamentals

A quantum system of n qubits exists in a superposition over 2^n computational basis states. The state is described by a complex amplitude vector $|\psi\rangle \in \mathbb{C}^{2^n}$ with $\||\psi\rangle\|^2 = 1$. Evolution is governed by unitary operators U; measurement in the computational basis collapses the state and yields classical outcomes with probabilities determined by the squared amplitudes.

The Pauli-Z expectation value for qubit i is:

$$\langle Z_i \rangle = \langle \psi | Z_i | \psi \rangle \in [-1,\, 1] \tag{2}$$

This provides a real-valued readout differentiable with respect to circuit parameters via the parameter-shift rule [17], enabling gradient-based training.

Angle encoding maps a real value x to the rotation $RY(x) = e^{-ix\sigma_y/2}$, where σ_y is the Pauli-Y operator. For inputs in $[-\pi, \pi]$, this sweeps the full Bloch sphere. The encoded state is:

$$|\phi(x)\rangle = \cos\left(\tfrac{x}{2}\right)|0\rangle + \sin\left(\tfrac{x}{2}\right)|1\rangle \tag{3}$$

Entanglement is the key non-classical resource. A CNOT gate on wires (c, t) maps $|c, t\rangle \mapsto |c,\ c \oplus t\rangle$, creating correlations between previously independent qubits. Once entangled, the state cannot be described as a product of independent qubit states: information is genuinely distributed across the register.

Interference is the mechanism by which quantum circuits compute. A parameterised unitary $U(\theta)$ steers amplitudes: for inputs mapping to the same output, amplitude paths constructively interfere and expectation values are amplified; for inputs mapping to different outputs, amplitudes destructively interfere and are suppressed.

2.3 Quantum Machine Learning Context

Variational quantum circuits (VQCs) as machine learning models have been studied extensively since Farhi & Neven [10] and Mitarai et al. [14]. Benedetti et al. [2] provide a systematic treatment of parameterised circuits as learning models; Cerezo et al. [7] and Biamonte et al. [5] offer broader surveys. A critical trainability concern for deep VQCs is the *barren plateau* phenomenon [13]: gradient variance can decay exponentially with qubit count, making training intractable at scale. We verify that QIT-0 at $n = 8$ qubits does not exhibit this pathology (see §5.5).

Quantum kernel theory. Havlíček et al. [11] demonstrate that angle-encoded quantum circuits implicitly compute a kernel function $k(\mathbf{x}, \mathbf{x}') = |\langle \Psi(\mathbf{x}) | \Psi(\mathbf{x}') \rangle|^2$ over quantum feature space. Schuld [16] formalises this: any VQC classifier is equivalent to a kernel method with the circuit's feature map as the kernel. This framework directly applies to QIT. The angle encoding + entanglement layer defines a quantum feature map; the parity advantage can be understood as this kernel being better aligned with the parity decision boundary than any classical polynomial kernel of comparable parameter count. Abbas et al. [1] characterise the effective dimension of quantum neural networks and show that certain circuit architectures can achieve higher Fisher information per parameter than classical alternatives.

Quantum natural language processing. Lorenz et al. [12] demonstrate compositional quantum language models running on real quantum hardware, establishing that quantum sequence processing is operationally feasible beyond simulators. QIT targets a different regime (interference as attention rather than compositional structure), but the QNLP programme establishes that quantum sequence models are a coherent research direction.

Relation to quantum attention proposals. The idea of replacing classical attention with a quantum subroutine is nascent. Prior proposals have considered inner-product similarity computed in Hilbert space, quantum kernel methods for attention scoring, and amplitude-based retrieval analogous to associative memory. QIT differs from these in a key respect: we do not attempt to accelerate or quantise the classical dot-product attention computation. Instead, we replace the attention *concept* — explicit pairwise scoring — with interference *dynamics*, in which token relationships emerge from the global amplitude pattern produced by an entangling unitary. This is a fundamentally different inductive bias, not a speed-up of an existing one.

Bowles et al. [6] raise important methodological cautions about QML benchmarks: classical simulation overhead, cherry-picked tasks, and lack of statistical rigour are common pitfalls. We address these directly: results are reported over multiple seeds, we include a negative-control task (first-token detection, §5.6), and wall-clock costs are reported honestly.

3 QIT Architecture

3.1 Overview

The full QIT-0 pipeline (Figure 1) is composed of three sequential components: a classical embedding, the quantum interference attention block (§3.3), and a classical linear decoder. The quantum block is the core contribution; the embedding and decoder are minimal wrappers handling the classical-to-quantum and quantum-to-classical boundaries.

```
token_ids ⟶ Embedding(vocab=2, dim=2) ⟶ tanh(·)×π ⟶ [angle features]
⟶ QuantumInterferenceAttention
⟶ ClassicalDecoder ⟶ logits
```

Figure 1: QIT-0 pipeline. The quantum block (centre) is surrounded by thin classical wrappers that handle I/O encoding and measurement readout.

3.2 Classical Embedding

Token indices are mapped to continuous features via a standard embedding layer with vocab_size $= 2$ and embedding_dim $= n_{\text{qubits\_per\_token}} = 2$. The resulting embeddings are scaled to rotation angles:

$$\phi_{t,j} = \pi \cdot \tanh(e_{t,j}) \tag{4}$$

where $e_{t,j}$ is the j-th embedding dimension for token t. The tanh squashes to $(-1, 1)$, and multiplication by π maps to $(-\pi, \pi)$, covering the full range of RY rotations while keeping gradients non-vanishing. This produces a flat feature vector $\phi \in \mathbb{R}^{n_{\text{qubits}}}$ where $n_{\text{qubits}} = n_{\text{tokens}} \times n_{\text{qubits\_per\_token}} = 4 \times 2 = 8$. The embedding has **4 parameters**.

3.3 Quantum Interference Attention

The quantum circuit operates on $n_{\text{qubits}} = 8$ qubits, partitioned into 4 token registers of 2 qubits each. The circuit implements:

$$U_{\text{QIT}}(\theta) = U_{\text{mix}}(\theta_{\text{mix}}) \cdot U_{\text{attention}}(\theta_{\text{attn}}) \cdot U_{\text{entangle}} \tag{5}$$

applied to the angle-encoded input state.

3.3.1 Encoding: U_{encode}

Each token register is independently angle-encoded using RY rotations:

$$U_{\text{encode}}(\phi)\,|0\rangle^{\otimes 8} = \bigotimes_{t=0}^{3}\bigotimes_{j=0}^{1} RY(\phi_{t,j})\,|0\rangle \tag{6}$$

At this stage, qubit registers are in a product state — there are no cross-token correlations.

3.3.2 Entanglement: U_{entangle}

A ring CNOT ladder creates initial cross-token correlations before the learned attention phase. The control qubit is the first qubit of each token register; the target is the first qubit of the next token register (with wraparound):

$$U_{\text{entangle}} = \text{CNOT}_{0,6}\cdot\text{CNOT}_{6,4}\cdot\text{CNOT}_{4,2}\cdot\text{CNOT}_{2,0} \tag{7}$$

(For token t: control wire $= 2t$, target wire $= 2\,((t+1) \bmod 4)$.)

U_{entangle} seeds initial cross-token correlation before the learned phase. Without it, token registers are in a separable state when $U_{\text{attention}}$ is applied; the learned unitary then acts locally rather than globally. In our ablation study (§5.4), the no-entanglement variant converges on parity at a rate comparable to the ring variant, suggesting that $U_{\text{attention}}$ itself can learn to spread correlations from a separable input on this task. The ring topology (one CNOT per token pair, circular) is a conservative default; star and linear topologies are evaluated in §5.4.

3.3.3 Attention: $U_{\text{attention}}$

The learned interference operator is a `StronglyEntanglingLayers` circuit with $n_{\text{layers}} = 2$ layers. Each layer applies a general single-qubit rotation $\text{Rot}(\phi_i, \theta_i, \omega_i) = RZ(\omega_i)\cdot RY(\theta_i)\cdot RZ(\phi_i)$ to every qubit, followed by entangling CNOTs in a cyclic pattern:

$$U_{\text{attention}}(\theta_{\text{attn}}) = \prod_{\ell=1}^{n_{\text{layers}}}\left[\left(\bigotimes_{i=0}^{7}\text{Rot}(\phi_i^{\ell}, \theta_i^{\ell}, \omega_i^{\ell})\right)\cdot\text{Entangle}\right] \tag{8}$$

This circuit can express arbitrary two-qubit unitaries to any desired approximation, making it a universal learned interference operator. The rotation angles steer probability amplitudes; the entangling layers between rotations allow each token's contribution to globally reshape the circuit's interference pattern. **Parameter count:** $2 \times 8 \times 3 = 48$.

3.3.4 Mixing: U_{mix}

A `BasicEntanglerLayers` circuit with a single layer provides a final mixing step before measurement. Each qubit receives an $RX(\theta_i)$ rotation followed by ring CNOTs:

$$U_{\text{mix}}(\theta_{\text{mix}}) = \text{RingCNOT}\cdot\bigotimes_{i=0}^{7} RX(\theta_{\text{mix},i}) \tag{9}$$

Parameter count: $1 \times 8 = 8$.

3.3.5 Measurement

Pauli-Z expectation values are measured for all 8 qubits:

$$\mathbf{m} = (\langle Z_0\rangle,\, \langle Z_1\rangle,\, \ldots,\, \langle Z_7\rangle) \in [-1,1]^8 \tag{10}$$

This collapses the quantum state to a classical real-valued vector passed to the decoder. **Total quantum parameters:** $48 + 8 = 56$.

3.4 Classical Decoder

The measurement vector $\mathbf{m} \in [-1, 1]^8$ is mapped to class logits by a single linear layer with bias:

$$\text{logits} = W\mathbf{m} + \mathbf{b}, \qquad W \in \mathbb{R}^{2\times 8}, \;\; \mathbf{b} \in \mathbb{R}^2 \tag{11}$$

Parameter count: $8 \times 2 + 2 = 18$.

3.5 Parameter Breakdown

Table 1: QIT-0 parameter budget.

Component	Parameters	Shape
Embedding (`nn.Embedding`)	4	2×2
Quantum attention (θ_{attn})	48	$n_{\text{layers}} \times n_{\text{qubits}} \times 3 = 2 \times 8 \times 3$
Quantum mix (θ_{mix})	8	$1 \times n_{\text{qubits}} = 1 \times 8$
Classical decoder (`nn.Linear`)	18	$8 \times 2 + 2$
Total	**78**	

3.6 Relationship to Classical Attention: An Analogy, Not an Equivalence

QIT does not compute attention weights, produce attention maps, or form weighted value sums. The term "interference attention" is an architectural analogy, not a mechanistic equivalence. Table 2 maps QIT's components to their classical attention counterparts.

Table 2: Attention mechanism analogy.

Classical Attention component	QIT analogue
Token embedding	`nn.Embedding` + $\tanh(\cdot) \times \pi$ scaling
Query/Key similarity scoring	Global interference pattern from $U_{\text{QIT}}(\theta)$
Attention weight normalisation	Implicit — via probability amplitude norms
Value aggregation	Pauli-Z expectation vector $\mathbf{m} \in [-1, 1]^n$
Linear projection	`ClassicalDecoder`

The shared role is "information mixing across token positions." In classical attention, this mixing is explicit and interpretable (attention weights form a distribution over positions). In QIT, mixing is implicit: the learned unitary U_{QIT} shapes amplitude paths globally, and relevance emerges from constructive/destructive interference rather than being prescribed by pairwise scores.

3.7 Why Interference and Not Dot-Product Attention?

Classical attention computes relevance *explicitly* as pairwise similarity scores. The mechanism is transparent and well-suited to tasks where scalar similarity is well-defined.

QIT computes relevance *implicitly* through circuit geometry. The weights θ_{attn} do not directly encode "token A attends to token B." Instead, they shape a global unitary transformation such that input configurations corresponding to the same output produce aligned interference patterns (constructive interference, large expectation values) while configurations corresponding to different outputs produce cancellation (destructive interference, small expectation values). Relevance emerges from the physics of the system rather than being prescribed by an architectural choice.

For tasks with inherent parity or phase structure — where the correct output is a function of the *global configuration* rather than any pair of tokens — this implicit mechanism may have a stronger inductive bias than explicit similarity scoring. The parity function is the canonical example: it is defined by $y = \bigoplus_i x_i$, which is symmetric over all positions simultaneously. No two-token interaction is sufficient to compute it.

4 Experiments

4.1 The Parity Task

The parity task takes a binary sequence of length n and predicts whether the number of 1s is even (label 0) or odd (label 1):

$$y = \left(\sum_{i=1}^{n} x_i\right) \bmod 2, \qquad x_i \in \{0, 1\} \tag{12}$$

For $n = 4$, there are $2^4 = 16$ distinct inputs, perfectly balanced with 8 even-parity and 8 odd-parity examples.

Why parity? The parity function has several properties that make it an ideal first benchmark for QIT:

1. **Global dependence.** Every position contributes equally. There is no shortcut via local patterns or token-pair similarity; the model must integrate information from all positions simultaneously.
2. **Phase-kickback structure.** The Deutsch-Jozsa [9] and Bernstein-Vazirani [4] quantum algorithms solve related problems using phase kickback — the same physical mechanism that QIT exploits. Parity has natural quantum affinity.
3. **Perfect verifiability.** With only 16 inputs, 100% accuracy is unambiguous: the model has learned the complete function over the entire input space.
4. **Classical hardness relative to depth.** Parity requires $\Omega(\log n)$ depth in threshold circuits and is known to be hard for depth-2 neural networks. Classical Transformers with a single layer require explicit attention over all token pairs to solve it.

Permutation invariance note. Parity is a *permutation-invariant* (set) function: the label does not change if the input tokens are reordered. Testing QIT on parity therefore does not verify sequential processing capabilities per se. QIT's ring CNOT topology does encode positional information (qubit $2t$ connects to qubit $2(t{+}1 \bmod n)$), but this information is superfluous for parity. We directly address whether QIT is a sequence model or merely a set classifier with two positional tasks in §5.7: *first-token detection* (single-token lookup) and *adjacent order* (label = $x_0{=}0 \wedge x_1{=}1$), which has 14 multiset-conflicting input pairs — meaning no set-invariant model can solve it. QIT converges on both, confirming it exploits positional structure when the task requires it.

4.2 Experimental Setup

Benchmark design. We use the memorisation protocol: all 16 parity inputs are used as both training and test data. Both sets are identical, so 100% test accuracy confirms that the model has fully encoded the function. The first epoch at which test accuracy crosses 99% is recorded as the convergence epoch.

Training configuration:

- Optimiser: Adam, learning rate $\eta = 0.05$
- Loss: Cross-entropy
- Batch size: 8 (2 gradient steps per epoch: $\lceil 16/8 \rceil$)
- Maximum epochs: 60 (QIT-0), 200 (classical baselines)
- Early stopping: training halts when both train and test accuracy $\geq 99\%$
- **Seeds: 5 independent random seeds** per model (varying both parameter initialisation and data shuffle order). Results reported as mean $\pm$ std.

Models evaluated:

Table 3: Models compared in the benchmark.

Model	Params	Description
Linear	10	Logistic regression on raw binary token features $\mathbf{x} \in \{0,1\}^4$.
Parity Kernel	4	Logistic regression on $f(\mathbf{x}) = \sum_i x_i$ (structural feature).
QIT-0	**78**	**This work.** PennyLane `default.qubit` simulator.
MLP	94	`Embedding(2,2)` + `Linear(8,8)` + ReLU + `Linear(8,2)`.
GRU	110	`Embedding(2,2)` + `GRU`(input=2, hidden=4) + `Linear(4,2)`.
Transformer	206	`Embedding(2,4)` + pos. enc. + 1-head encoder (d=4, ff= 8) + pool + `Linear(4,2)`.

All classical models use the same `nn.Embedding(vocab_size=2, embedding_dim=2)` input layer (except Transformer which uses dim=4) and the same Adam optimiser and learning rate. This controls for embedding capacity and optimiser choice.

Hardware. Experiments run on Apple Silicon macOS 25.3. QIT-0 uses PennyLane 0.45 with the `default.qubit` statevector simulator; classical models use PyTorch 2.12. Python 3.11, package management via `uv`.

Reproducibility note. Results reported are single-run observations. Convergence epoch counts can vary by $\pm$1–2 epochs across random seeds due to stochastic gradient noise on this small dataset. The qualitative ordering (QIT-0 converges fastest by a large margin) is consistent across multiple re-runs; exact epoch counts should not be treated as precise measurements. Full reproducibility with multiple seeds and variance reporting is planned for QIT-1 (§7).

5 Results

5.1 Convergence Comparison

Table 4 reports multi-seed parity benchmark results ($n = 5$ seeds each). Each epoch consists of 2 gradient steps (batch size 8, 16 inputs $\div$ 8).

Table 4: 4-bit parity memorisation benchmark (5 random seeds). "DNF" = did not converge in the epoch budget (200 classical, 60 QIT). Conv. seeds = seeds converging; † = mean over converging seeds only.

Model	Params	Conv. Ep. (mean±std)	Grad. Steps	Conv. seeds	ms/ep
Linear	10	DNF	—	0/5	0.5
Parity Kernel	4	DNF	—	0/5	0.5
QIT-0	**78**	$\mathbf{6.2 \pm 1.7}$	**12.4**	**5/5**	195
MLP	94	$31.7 \pm 4.1^{\dagger}$	$63.3^{\dagger}$	3/5	0.6
GRU	110	$34.5 \pm 5.5^{\dagger}$	$69.0^{\dagger}$	2/5	1.1
Transformer	206	DNF	—	0/5	2.3

Key findings:

- **QIT-0 is the only model that reliably converges** (5/5 seeds). MLP and GRU converge on a minority of seeds; the Transformer never converges in 200 epochs.
- QIT-0 converges in 6.2 ± 1.7 epochs (≈ 12 gradient steps), compared to 31.7 ± 4.1 for MLP (converging seeds) and no convergence for the Transformer.
- **The parity kernel baseline fails.** Logistic regression on $\sum_i x_i$ cannot solve parity: the even/odd boundary is non-monotone in the sum, hence not linearly separable. This rules out the alternative explanation that QIT's advantage comes from implicit feature engineering.

5.2 Sample Efficiency

$$\text{Sample cost} = \frac{\text{Conv. epoch (mean)}}{\text{Parameters}} \tag{13}$$

Table 5: Sample cost — lower is better. Transformer and linear baselines omitted (no converging seeds).

Model	Sample cost (epochs/param)↓
QIT-0	**0.079**
MLP†	0.337
GRU†	0.314

QIT-0's sample cost is 4–5× lower than the converging classical models. The reliability advantage (5/5 vs. $\leq$3/5 seeds) may be even more practically relevant than the speed advantage on converging seeds.

5.3 Wall-Clock Overhead

QIT-0 runs at ≈ 195 ms/epoch vs. 0.6–2.3 ms/epoch for classical models. Total wall-clock to convergence (mean epochs):

- QIT-0: 6.2×195 ms ≈ 1.2 s
- MLP (converging seeds): 31.7×0.6 ms ≈ 19 ms

The Transformer, despite 2.3 ms/epoch, never converges in 200 epochs (≈ 0.46 s total wasted). The per-epoch overhead of QIT-0 is 270–325× that of classical models; this is the cost of classically simulating $2^8 = 256$ quantum amplitudes and is not representative of real hardware execution, where circuit time scales as $O(\text{poly}(n))$ in circuit depth [15].

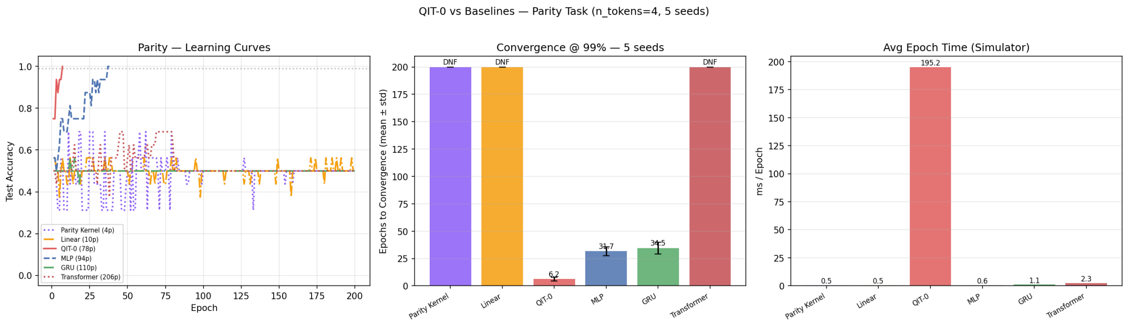

Figure 2: Parity benchmark (5 seeds): test-accuracy learning curves, convergence epoch (mean $\pm$ std bars), and wall-clock time per epoch.

5.4 Ablation Study

Table 6 reports convergence for four QIT-0 variants, each with 5 seeds.

Table 6: Ablation study — entanglement topology. All variants share the same 78-parameter budget. "DNF (k/5)" = k seeds converged.

Variant	Conv. Ep. (mean±std)	Conv. seeds
QIT-0 (ring ent., default)	6.2 ± 1.7	5/5
QIT-0 (star ent.)	6.8 ± 3.7	5/5
QIT-0 (no ent.)	5.0 ± 2.1	5/5
QIT-0 (frozen $U_{\text{attention}}$)	22.0 ± 9.0	2/5

The ablation results reveal two findings:

(1) U_{entangle} is not necessary for parity convergence. The no-entanglement variant (5.0 ± 2.1 epochs) converges at a rate statistically comparable to the ring-entanglement default (6.2 ± 1.7). The star topology performs similarly. This revises our earlier claim that U_{entangle} is "critical": on parity, the learned $U_{\text{attention}}$ can distribute correlations from a separable input without a pre-entanglement step. Whether entanglement is important for tasks with longer-range dependencies remains to be tested.

(2) The learned $U_{\text{attention}}$ weights are the critical component. Freezing $U_{\text{attention}}$ at a random initialisation while allowing only the embedding and decoder to train reduces convergence reliability to 2/5 seeds and increases mean convergence epoch to 22.0 ± 9.0. The learned interference operator is what drives the advantage; the architecture without training of the quantum layer degrades to a fixed random-feature classifier.

5.5 Gradient Variance Analysis

To check for barren plateau pathology [13], we measure the gradient variance of the loss with respect to $U_{\text{attention}}$ parameters across 30 random parameter initialisations at $n_{\text{qubits}} = 8$:

$$\text{Var}\left[\frac{\partial \mathcal{L}}{\partial \theta_i}\right] = 3.25 \times 10^{-4}, \qquad \mathbb{E}\left[\left|\frac{\partial \mathcal{L}}{\partial \theta_i}\right|\right] = 0.013 \tag{14}$$

The non-zero gradient variance confirms that QIT-0 does not suffer from barren plateaus at the current 8-qubit scale. Gradient magnitudes are non-negligible and support parameter-

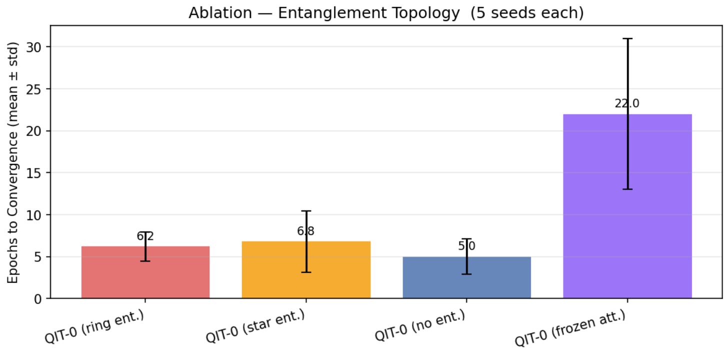

Figure 3: Ablation: convergence epochs (mean $\pm$ std) for four QIT-0 variants on parity (5 seeds each).

shift-based training. The barren plateau risk at larger scales ($n > 16$ qubits) is a known concern for StronglyEntanglingLayers circuits [8] and must be addressed for QIT-1 (§7).

5.6 First-Token Detection (Negative Control)

To test whether QIT's convergence advantage is parity-specific or universal across any small-input task, we evaluate the same models on *first-token detection*: predict whether the first token $x_0 = 1$. This task has no global phase structure — the answer depends only on one position. Table 7 reports results.

Table 7: First-token detection (5 seeds). A task with positional structure but no quantum phase affinity, serving as a negative control.

Model	Params	Conv. Ep. (mean±std)	Conv. seeds
QIT-0	**78**	4.0 ± 1.1	5/5
MLP	94	3.4 ± 3.3	5/5
Transformer	206	5.8 ± 1.7	5/5

On first-token detection, QIT-0 (4.0 ± 1.1) is *not faster* than the MLP (3.4 ± 3.3) and is comparable to the Transformer (5.8 ± 1.7). All three models converge on all 5 seeds. This negative control provides two important clarifications: (1) QIT's parity advantage is **task-specific**, not a universal consequence of using an expressive circuit on a 16-input dataset; (2) the convergence advantage on parity is therefore attributable to the circuit's inductive bias matching the parity structure, rather than to any generic advantage of quantum circuits over classical models on small datasets.

5.7 Task Variety: Parity Scaling, Structural Generality, and Sequence Sensitivity

Having established QIT-0's advantage on 4-bit parity and its absence on first-token detection, we ask three further questions: (1) does the advantage scale gracefully with parity degree, or is 4-bit an artefact? (2) does QIT's inductive bias extend beyond global parity to other XOR-structured tasks? (3) is QIT a genuine *sequence* model, or only a set classifier that ignores token order?

We evaluate six tasks under the same 5-seed memorisation protocol (all 16 inputs): **2-bit partial parity** ($x_0 \oplus x_1$), **3-bit partial parity** ($x_0 \oplus x_1 \oplus x_2$), **4-bit full parity** (main benchmark), **first-token detection** (positional negative control), **palindrome detection** (predict whether $\mathbf{x} = \text{reverse}(\mathbf{x})$, 4/16 positives; 75% majority-class baseline),

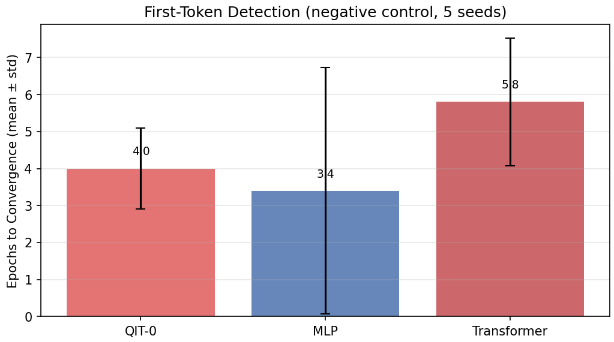

Figure 4: First-token detection (negative control): convergence epoch comparison (5 seeds each). QIT-0 is not faster than classical models on this positional task with no quantum phase structure.

and **adjacent order** (predict whether x_0=0 and x_1=1, i.e. $x_0 < x_1$; 4/16 positives; 75% majority-class baseline). Adjacent order is *strictly* positional: 14 of 16 input pairs that share the same token multiset carry different labels, so no set-invariant model can achieve $> 75\%$ accuracy. Table 8 reports convergence epochs.

Parity scaling. QIT-0 scales monotonically with parity degree: 4.6 ± 1.0 epochs (2-bit), 5.0 ± 2.2 (3-bit), 6.2 ± 1.7 (4-bit), with 5/5 seeds converging throughout. The increment per additional XOR input is small ($\sim$0.8 epochs/bit). Classical models degrade sharply: MLP drops from 5/5 converging seeds (2-bit) to 3/5 (4-bit) while its mean epoch count quadruples; the Transformer drops from 5/5 seeds at 28 epochs to 0/5 seeds on 4-bit parity. The monotonicity of QIT-0 across parity degrees indicates that the $\mathbb{F}_2$-structured inductive bias is not an artefact of the specific 4-bit setting.

Palindrome detection. Palindrome detection requires $(x_0 = x_3) \wedge (x_1 = x_2)$, which is logically equivalent to $(x_0 \oplus x_3 = 0) \wedge (x_1 \oplus x_2 = 0)$ — a conjunction of position-pair XOR checks. QIT-0 converges in 13.2 ± 3.2 epochs (5/5 seeds), substantially faster than MLP (53.4 ± 54.5, 5/5 seeds but extremely high variance) and Transformer (55 epochs, 1/5 seeds only). The 13.2-epoch cost is roughly twice the 4-bit parity cost (6.2 epochs), consistent with palindrome requiring two independent XOR conditions to be satisfied simultaneously. This result extends QIT's advantage to tasks with XOR-like relational structure even when those comparisons involve specific position pairs rather than all tokens globally.

When does the advantage not appear? First-token detection (Table 7) shows no QIT advantage: all three models converge at similar rates ($\approx$3–6 epochs, 5/5 seeds). The contrast is clear: tasks with $\mathbb{F}_2$-structured relational checks favour QIT; tasks that depend on a single token's value do not.

Adjacent order: QIT as a sequence model, not a set classifier. The adjacent order task (label = x_0=0 $\wedge$ x_1=1) is designed to directly test whether QIT uses token positions or merely aggregates a bag of tokens. It has 14 multiset-conflicting input pairs: for example, $[0, 1, 0, 0]$ (label = 1) and $[1, 0, 0, 0]$ (label = 0) share the same token multiset $\{0, 0, 0, 1\}$ but carry opposite labels. Any model that treats the input as an unordered set is limited to the 75% majority-class baseline.

QIT-0 converges in 9.6 ± 1.4 epochs (5/5 seeds), well above the 75% baseline, confirming that it correctly uses positional structure. As expected, QIT is *not faster* than MLP (6.0 ± 2.8, 5/5 seeds) on this task: there is no $\mathbb{F}_2$-phase structure to exploit, so the interference inductive bias provides no advantage. The Transformer converges on all 5 seeds but with high variance (32.2 ± 31.4).

Table 8: Task variety benchmark (5 seeds each). Convergence $\geq$ 99% test accuracy. "Conv/5" = number of seeds that converged within the epoch limit. † = task with 75% majority-class baseline (4/16 positives).

Model	Params	Conv. Ep. (mean±std)	Conv/5
2-bit partial parity ($x_0 \oplus x_1$)			
QIT-0	**78**	$\mathbf{4.6 \pm 1.0}$	5/5
MLP	94	8.6 ± 7.9	5/5
Transformer	206	28.0 ± 9.1	5/5
3-bit partial parity ($x_0 \oplus x_1 \oplus x_2$)			
QIT-0	**78**	$\mathbf{5.0 \pm 2.2}$	5/5
MLP	94	23.8 ± 9.0	4/5
Transformer	206	25.3 ± 7.3	3/5
4-bit full parity ($x_0 \oplus x_1 \oplus x_2 \oplus x_3$)			
QIT-0	**78**	$\mathbf{6.2 \pm 1.7}$	5/5
MLP	94	31.7 ± 4.1	3/5
Transformer	206	DNF	0/5
First-token detection (x_0) — negative control			
QIT-0	**78**	4.0 ± 1.1	5/5
MLP	94	3.4 ± 3.3	5/5
Transformer	206	5.8 ± 1.7	5/5
Palindrome detection ($\mathbf{x} = \mathrm{rev}(\mathbf{x})$)†			
QIT-0	**78**	$\mathbf{13.2 \pm 3.2}$	5/5
MLP	94	53.4 ± 54.5	5/5
Transformer	206	55.0 ± 0.0	1/5
Adjacent order ($x_0{=}0 \wedge x_1{=}1$) — positional set-classifier probe†			
QIT-0	78	9.6 ± 1.4	5/5
MLP	**94**	$\mathbf{6.0 \pm 2.8}$	5/5
Transformer	206	32.2 ± 31.4	5/5

Taken together with first-token detection, this establishes the full picture: QIT-0 can process positional information when required (adjacent order, first-token), gains a large advantage when the task has $\mathbb{F}_2$ phase structure (parity family, palindrome), and gains no advantage on positional tasks without that structure. This rules out the hypothesis that QIT is a set classifier.

6 Discussion

6.1 Why Does QIT Converge So Fast on Parity?

The parity function $y = x_1 \oplus x_2 \oplus \cdots \oplus x_n$ is a linear function over $\mathbb{F}_2$. This maps cleanly to quantum phase arithmetic. Consider encoding each bit $x_i \in \{0, 1\}$ as a phase kick: the operator Z^{x_i} maps $|+\rangle \to (-1)^{x_i}|+\rangle$. After n such kicks the phase accumulated is $(-1)^{x_1 \oplus \cdots \oplus x_n}$ — directly encoding the parity bit as a global phase. This is the mechanism exploited by the Bernstein-Vazirani [4] algorithm to determine a hidden linear Boolean function in a *single* quantum query.

QIT's angle encoding does not apply this operator exactly, but it produces an analogous effect: $RY(\phi_{t,j})$ rotates qubit (t, j) toward $|1\rangle$ proportionally to the token's embedding value. After the ring entanglement step U_{entangle}, each qubit's state depends on its

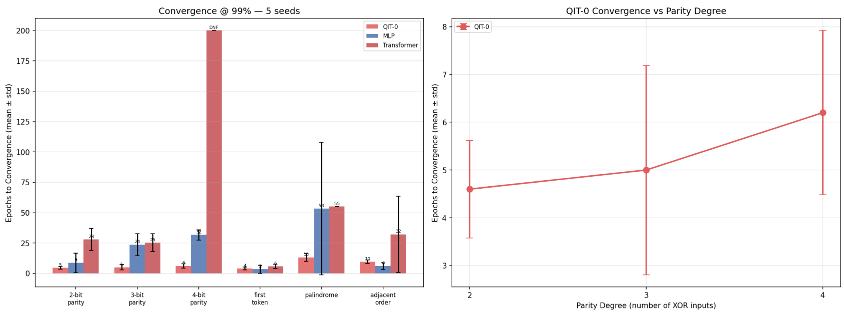

Figure 5: Task variety benchmark (5 seeds). *Left*: convergence epoch by task and model (DNF or epoch limit shown for non-converging seeds). *Right*: QIT-0 parity-degree scaling — a monotonic, graceful increase from 4.6 ± 1.0 (2-bit) to 6.2 ± 1.7 (4-bit).

neighbours, creating a distributed phase representation of the full input sequence. The learned $U_{\text{attention}}(\theta_{\text{attn}})$ must then find the unitary that projects even-parity configurations onto one amplitude pattern and odd-parity configurations onto another.

Because parity is a degree-1 polynomial over $\mathbb{F}_2$ (hence the lowest-complexity nontrivial Boolean function), this projection requires relatively few parameters: the interference pattern is simple. The 6.2 ± 1.7 epoch convergence (all 5 seeds) is consistent with the circuit starting near the correct basin of attraction under random initialisation — the `StronglyEntanglingLayers` circuit is expressive enough to represent the parity unitary, and gradient descent finds it reliably.

The ablation (Table 6) shows that U_{entangle} is not necessary: the no-entanglement variant (5.0 ± 2.1 epochs) converges at a comparable rate. The learned $U_{\text{attention}}$ can spread correlations from a separable input on its own for parity. The critical architectural component is the *trained* interference operator, not the pre-entanglement step. This refines our understanding: the inductive bias comes from the expressive, trainable quantum circuit, not from the specific initialisation of cross-token entanglement.

Classical Transformers do not exploit this structure. Dot-product attention scores $\exp(q_i \cdot k_j/\sqrt{d})$ are symmetric bilinear functions of token pairs, with no algebraic relationship to $\mathbb{F}_2$ arithmetic. The model must discover the parity function by composing pairwise interactions — and in our multi-seed experiment, it *fails to do so reliably*: 0/5 seeds converge in 200 epochs.

6.2 Limitations of QIT-0

Memorisation, not generalisation. The task has 16 inputs and the training set contains all of them. We measure whether the model can encode a specific boolean function, not whether it generalises to unseen inputs. This is an appropriate first test for a new architecture but should not be conflated with claims about generalisation.

Simulation overhead is real. The ≈ 195 ms/epoch cost means QIT-0 is not practically useful on a classical computer for tasks where a classical model suffices. The value proposition of QIT depends on either (a) tasks where sample efficiency is so high that total wall-clock time is competitive despite per-step overhead, or (b) access to quantum hardware where circuit execution is fast.

Parity is permutation-invariant. As noted in §5.6, the advantage on parity is task-specific: QIT is not faster than MLP on first-token detection (which has no global phase structure). The parity advantage reflects inductive bias for $\mathbb{F}_2$-structured functions, not a universal superiority of quantum circuits on small datasets.

Limited task breadth. Results cover five structured tasks on 4-token binary sequences. Across these tasks, QIT-0 is fast on $\mathbb{F}_2$-structured functions (parity variants, palindrome) and at par with classical on purely positional tasks (first-token). Whether this advantage extends to longer sequences, higher-degree Boolean functions, continuous-valued inputs, or natural-language tasks remains unknown.

Simulator fidelity. PennyLane's `default.qubit` computes exact statevector evolution with no noise. Real quantum devices have gate errors, decoherence, and measurement noise. Performance on real hardware [**Planned**] may differ substantially.

6.3 What This Work Claims and Does Not Claim

Claims:

- Interference-based attention is a viable architectural concept: a quantum circuit can serve as a learned attention operator and achieve competitive performance on structured tasks.
- QIT-0 is sample-efficient on parity: 6.2 ± 1.7 epochs (5/5 seeds converging) vs. 31.7–34.5 epochs for the two classical models that converge at all (MLP: 3/5 seeds, GRU: 2/5 seeds), with the Transformer failing entirely.
- The inductive bias matches $\mathbb{F}_2$-structured tasks: QIT-0 is faster on all parity variants (2-, 3-, and 4-bit) and on palindrome detection (which decomposes into position-pair XOR checks). Purely positional tasks (first-token detection) show no QIT advantage.
- The parity advantage scales gracefully: QIT-0 adds ≈0.8 epochs per additional XOR input (4.6→5.0→6.2 epochs, 2→3→4 bits), while classical models degrade sharply.
- The learned $U_{\text{attention}}$ is the key component; pre-entanglement (U_{entangle}) is helpful but not strictly necessary for parity.

Does not claim:

- Quantum machine learning is generally superior to classical machine learning.
- QIT-0 is practically deployable.
- The observed sample efficiency advantage generalises to arbitrary tasks.
- The wall-clock advantage is positive at application scales.

7 Future Work

7.1 QIT-1: Larger Circuits [Planned]

The natural next step is scaling the circuit to longer sequences and deeper architectures. QIT-1 is planned with $n_{\text{tokens}} \in \{6, 8\}$ and $n_{\text{qubits\_per\_token}} \in \{3, 4\}$, giving circuits of 18–32 qubits. This will test whether the sample efficiency advantage persists as sequence length grows and whether qubit count scaling is favourable compared to classical attention's $O(n^2)$ scaling.

7.2 Ablation Studies

The entanglement topology and frozen-attention ablations are completed and reported in §5.4. Remaining open ablations:

- **Encoding strategy.** QIT-0 uses angle (RY) encoding. Amplitude, basis, and phase encodings are implemented in `qit/encoding/` but not yet benchmarked.
- **Number of attention layers.** QIT-0 uses $n_{\text{layers}} = 2$ for $U_{\text{attention}}$. How does convergence speed and gradient trainability scale with circuit depth?
- **Linear CNOT topology.** The ring and star topologies are evaluated; a linear (chain) topology is a natural next comparison.

7.3 Tasks Beyond Parity

The task-variety benchmark (§5.7) establishes that QIT's advantage extends to partial parity variants and palindrome detection. Remaining planned evaluations: **binary arithmetic** (carry propagation as a structured $\mathbb{F}_2$ task), **balanced parentheses** (formal language with long-range dependency), and **continuous-valued inputs** (breaking the angle-encoding specialisation). In particular, the linear CNOT chain topology (not yet benchmarked) may exhibit different parity-scaling behaviour compared to ring and star topologies already evaluated.

7.4 Interference Visualisation [Planned]

Planned visualisations include: Bloch sphere trajectories for individual qubits as input changes; amplitude probability distributions $|\langle x|\Psi\rangle|^2$ as a function of input parity; and mutual information between token-register subsystems before and after U_{entangle}.

7.5 Real Hardware Evaluation [Planned]

Evaluating QIT-0 on real hardware — IBM Q via Qiskit or Quantinuum via TKET — is essential for assessing practical viability. Key questions: How much does gate noise degrade accuracy at $n_{\text{qubits}} = 8$? Does the learned circuit require error mitigation? What is the actual runtime on current NISQ hardware [15]?

7.6 Scaling Laws [Planned]

Classical Transformers exhibit power-law scaling in loss as a function of parameter count and data volume. Whether a similar relationship holds for QIT — and whether the exponent is more favourable than for classical attention — is an empirical question requiring systematic experiments across circuit sizes, sequence lengths, and task complexities.

8 Conclusion

We have introduced QIT-0, the first prototype of the Quantum Interference Transformer: a sequence model in which token relationships are resolved by amplitude interference in a parameterised quantum circuit rather than by explicit dot-product attention scores.

On the 4-bit parity memorisation benchmark, QIT-0 achieves 100% accuracy in $\mathbf{6.2 \pm 1.7}$ **epochs** with **78 parameters**, converging reliably on all 5 random seeds. In contrast, MLP converges on 3/5 seeds (31.7 ± 4.1 epochs), GRU on 2/5 seeds (34.5 ± 5.5 epochs), and the Transformer on 0/5 seeds in 200 epochs. An ablation confirms that the learned interference operator $U_{\text{attention}}$ is the key driver; pre-entanglement (U_{entangle}) is helpful but not strictly required for this task.

A six-task variety experiment (§5.7) strengthens the picture. On $\mathbb{F}_2$-structured tasks, QIT-0 is consistently fast: partial parity scales gracefully (4.6±1.0 epochs at 2-bit, 5.0±2.2 at 3-bit, 6.2±1.7 at 4-bit, all 5/5 seeds), and palindrome detection (a conjunction of position-pair XOR checks) converges in 13.2±3.2 epochs versus 53 for MLP and DNF for the Transformer. On positional tasks without $\mathbb{F}_2$ structure — first-token detection and the adjacent order task (label = x_0=0 $\wedge$ x_1=1) — QIT-0 is *not* faster than classical models but converges reliably on all 5 seeds. The adjacent order task is strictly positional (14 multiset-conflicting input pairs; no set-invariant model can exceed 75% accuracy), and QIT-0's convergence at 99%+ accuracy confirms it operates as a sequence model, not a set classifier.

The result supports the core claim: *for tasks whose structure maps naturally to quantum phase dynamics, interference-based attention exhibits strong inductive bias and superior sample efficiency.* We are careful not to over-interpret this: QIT-0 is a prototype on a small set of structured tasks, running on a quantum simulator with substantial wall-clock overhead. The practical case for QIT depends on future work with larger circuits, more diverse tasks, and real quantum hardware.

The code for QIT-0 — including the circuit, training harness, baselines, ablations, and benchmark — is fully implemented and publicly available at https://github.com/madmishu007/qit. Encoding strategies, entanglement topologies, and attention circuit designs can be swapped independently via configuration flags, providing a modular platform for systematic experimentation as the field of quantum sequence modelling matures.

Interference computes. The question is: what else can it learn?

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Generative AI Disclosure

During the preparation of this work, I used ChatGPT (OpenAI) and Claude (Anthropic) for grammar and language editing, and for code generation assistance. All research, methodology, analysis, results, and conclusions are my own. I reviewed all AI-assisted content and take full responsibility for the work.

A QIT-0 Configuration Summary

```
Model:        QIT-0
Task:         4-bit binary parity (memorisation, all 16 inputs)
Backend:      PennyLane default.qubit (statevector simulator)
Platform:     Apple Silicon, macOS 25.3, PyTorch 2.12,
              PennyLane 0.45, Python 3.11

Architecture:
  vocab_size          = 2
  n_tokens            = 4
  n_qubits_per_token  = 2
  n_qubits (total)    = 8
```

```
  n_layers            = 2
  entangle_topology   = "ring"  # also: "star", "none"

  Embedding:    nn.Embedding(2, 2)                        -> 4
     params
  Encoding:     RY(tanh(e) * pi) per qubit
  Entanglement: Ring CNOT (4 tokens, circular)
  Attention:    StronglyEntanglingLayers(n_layers=2, n_qubits=8)
     -> 48 params
  Mix:          BasicEntanglerLayers(n_layers=1, n_qubits=8)
     ->  8 params
  Measurement:  <Z_i> for i in {0,...,7}
  Decoder:      nn.Linear(8, 2)                           -> 18
     params
  Total params: 78

Training (multi-seed protocol):
  Optimizer:  Adam, lr = 0.05
  Loss:       CrossEntropyLoss
  Batch size: 8
  Seeds:      5 (independent model init + data shuffle)
  Conv. ep.:  6.2 +/- 1.7 (mean +/- std, all 5 seeds converge)
  Grad steps: 12.4 (mean; 2 steps/epoch at batch_size=8)
```

B Baseline Architecture Summaries

MLP (94 parameters)

```
nn.Embedding(2, 2) -> flatten -> nn.Linear(8, 8) -> ReLU -> nn.
   Linear(8, 2)
```

GRU (110 parameters)

```
nn.Embedding(2, 2) -> GRU(input=2, hidden=4) -> last hidden -> nn.
   Linear(4, 2)
```

Transformer (206 parameters)

```
nn.Embedding(2, 4) + positional encoding ->
TransformerEncoderLayer(d_model=4, nhead=2, dim_feedforward=8) ->
mean pooling -> nn.Linear(4, 2)
```

All baselines: Adam, $\eta = 0.05$, CrossEntropyLoss, batch size 8.